

EXPLORATORY DATA ANALYSIS USING POWER BI

WEEK – 5

REPORT

HEALTHCARE DATA CLEANING AND DATA MODELING

SUBMITTED BY

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E25DA014

Introduction:

The healthcare dataset was cleaned and transformed using Power BI. The cleaned data was then organized into dimension and fact tables to create a proper data model for analysis and reporting.

Objectives:

- To clean and transform the healthcare dataset.
- To standardize text and data types.
- To create an **Age Group** column using conditional logic.
- To create dimension and fact tables using references.
- To merge required columns and create relationships between tables.
- To prepare the dataset for effective analysis in Power BI.

Dataset Used:

‘Healthcare_dataset.csv’

Data Cleaning:

The healthcare dataset was cleaned using **Power Query** in Power BI.

The following transformations were performed:

1. Trimmed Text

- Unwanted spaces from text fields were removed using the Trim operation.
- This helped maintain consistency in the data.

2. Changed Data Types

- Appropriate data types were assigned to the columns.
- For example, age and billing amount were treated as numeric values, while admission and discharge dates were changed to date format.

3. Capitalized Each Word

- Text values were formatted using the **Capitalize Each Word** option.
- This made names and other text-based fields more consistent and readable.

Query Settings

Table.Distinct(#"Removed Duplicates", {"Name"})

	A ₁ Name	A ₂ Age	A ₃ Gender	A ₄ Blood Type	A ₅ Medical Condition	A ₆ Date of Admission	A ₇ Doctor
1	Bobby Jackson	30	Male	B-	Cancer	1/31/2024	Matthew Smit
2	Leslie Terry	62	Male	A+	Obesity	8/20/2019	Samantha D
3	Danny Smith	76	Female	A-	Obesity	9/22/2022	Tiffany Mitch
4	Andrew Watts	28	Female	O+	Diabetes	11/18/2020	Kevin Wells
5	Adrienne Bell	43	Female	AB+	Cancer	9/19/2022	Kathleen Han
6	Emily Johnson	36	Male	A+	Asthma	12/20/2023	Taylor Newto
7	Edward Edwards	21	Female	AB-	Diabetes	11/3/2020	Kelly Olson
8	Christina Martinez	20	Female	A+	Cancer	12/28/2021	Suzanne Thor
9	Jasmine Aguilar	82	Male	AB+	Asthma	7/1/2020	Daniel Fergus
10	Christopher Berg	58	Female	AB-	Cancer	5/23/2021	Heather Day
11	Michelle Daniels	72	Male	O+	Cancer	4/19/2020	John Duncan
12	Aaron Martinez	38	Female	A-	Hypertension	8/13/2023	Douglas May
13	Connor Hansen	75	Female	A+	Diabetes	12/12/2019	Kenneth Fletc
14	Robert Bauer	68	Female	AB+	Asthma	5/22/2020	Theresa Free
15	Brooke Brady	44	Female	AB+	Cancer	10/8/2021	Roberta Stew
16	Ms. Natalie Gamble	46	Female	AB-	Obesity	1/1/2023	Maria Dought
17	Haley Perkins	63	Female	A+	Arthritis	6/23/2020	Erica Spencer
18	Mrs. Jamie Campbell	38	Male	AB-	Obesity	3/8/2020	Justin Kim
19	Luke Burgess	34	Female	A-	Hypertension	3/4/2021	Justin Moore
20	Daniel Schmidt	63	Male	B+	Asthma	11/15/2022	Denise Gallov
21	Timothy Burns	67	Female	A-	Asthma	6/28/2023	Krista Smith
22	Christopher Bright	48	Male	B+	Asthma	1/21/2020	Gregory Smit
23	Kathryn Stewart	58	Female	O+	Arthritis	5/12/2022	Vanessa New
24	Dr. Eileen Thompson	59	Male	A+	Asthma	8/2/2021	Donna Martir
25	Paul Henderson	72	Female	AB+	Hypertension	5/15/2020	Stephanie Krc
26	Cathy Small	51	Female	O-	Asthma	12/23/2023	Wendy Glenn
27	Mr. Kenneth Moore	34	Female	A+	Diabetes	6/21/2022	James Ellis
28	Mary Hunter	38	Female	O-	Hypertension	1/3/2021	Jared Bruce J
29	Joshua Oliver	63	Female	B+	Hypertension	10/3/2023	Brandy Mitch
30	Thomas Martinez	34	Male	B-	Asthma	8/18/2019	Jacob Huynh
31	James Patterson	23	Female	A+	Arthritis	11/3/2019	Kristina Frazi

Query Settings

PROPERTIES

Name

healthcare_dataset

APPLIED STEPS

Source

Promoted Headers

Changed Type

Capitalized Each Word

Trimmed Text

Changed Type with Locale

Added Conditional Column

Changed Type with Locale1

Removed Duplicates

Removed Duplicates1

4. Created Age Group

- A conditional column named **Age Group** was created.
- Patients were categorized into different age groups based on their age.
- This makes it easier to analyze patients according to age categories.

Query Settings

Table.TransformColumnTypes(#"Trimmed Text", ({"Discharge Date", type date}), "en-US")

	A ₁ Name	A ₂ Age	A ₃ Gender	A ₄ Blood Type	A ₅ Medical Condition	A ₆ Date of Admission	A ₇ Doctor
1	Bobby Jackson	30	Male	B-	Cancer	1/31/2024	Matthew Smit
2	Leslie Terry	62	Male	A+	Obesity	8/20/2019	Samantha D
3	Danny Smith	76	Female	A-	Obesity	9/22/2022	Tiffany Mitch
4	Andrew Watts	28	Female	O+	Diabetes	11/18/2020	Kevin Wells
5	Adrienne Bell	43	Female	AB+	Cancer	9/19/2022	Kathleen Han
6	Emily Johnson	36	Male	A+	Asthma	12/20/2023	Taylor Newto
7	Edward Edwards	21	Female	AB-	Diabetes	11/3/2020	Kelly Olson
8	Christina Martinez	20	Female	A+	Cancer	12/28/2021	Suzanne Thor
9	Jasmine Aguilar	82	Male	AB+	Asthma	7/1/2020	Daniel Fergus
10	Christopher Berg	58	Female	AB-	Cancer	5/23/2021	Heather Day
11	Michelle Daniels	72	Male	O+	Cancer	4/19/2020	John Duncan
12	Aaron Martinez	38	Female	A-	Hypertension	8/13/2023	Douglas May
13	Connor Hansen	75	Female	A+	Diabetes	12/12/2019	Kenneth Fletc
14	Robert Bauer	68	Female	AB+	Asthma	5/22/2020	Theresa Free
15	Brooke Brady	44	Female	AB+	Cancer	10/8/2021	Roberta Stew
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17	Haley Perkins	63	Female	A+	Arthritis	6/23/2020	Erica Spencer
18	Mrs. Jamie Campbell	38	Male	AB-	Obesity	3/8/2020	Justin Kim
19	Luke Burgess	34	Female	A-	Hypertension	3/4/2021	Justin Moore
20	Daniel Schmidt	63	Male	B+	Asthma	11/15/2022	Denise Gallov
21	Timothy Burns	67	Female	A-	Asthma	6/28/2023	Krista Smith
22	Christopher Bright	48	Male	B+	Asthma	1/21/2020	Gregory Smit
23	Kathryn Stewart	58	Female	O+	Arthritis	5/12/2022	Vanessa New
24	Dr. Eileen Thompson	59	Male	A+	Asthma	8/2/2021	Donna Martir
25	Paul Henderson	72	Female	AB+	Hypertension	5/15/2020	Stephanie Krc
26	Peter Fitzgerald	73	Male	AB+	Obesity	5/15/2020	Angela Contr
27	Cathy Small	51	Female	O-	Asthma	12/23/2023	Wendy Glenn
28	Mr. Kenneth Moore	34	Female	A+	Diabetes	6/21/2022	James Ellis
29	Mary Hunter	38	Female	O-	Hypertension	1/3/2021	Jared Bruce J
30	Joshua Oliver	63	Female	B+	Hypertension	10/3/2023	Brandy Mitch
31	Thomas Martinez	34	Male	B-	Asthma	8/18/2019	Jacob Huynh

Query Settings

PROPERTIES

Name

healthcare_dataset

APPLIED STEPS

Source

Promoted Headers

Changed Type

Capitalized Each Word

Trimmed Text

Changed Type with Locale

fx

Table.Distinct("#Removed Duplicates", {"Name"})

nn profiling based on top 1000 rows

Age group were added in Healthcare_dataset Table

Data Modeling:

After cleaning the original healthcare dataset, references were created to organize the data into separate tables.

The main tables created were:

1. Dim Patient

The **Dim Patient** table contains patient-related information.

Relevant fields include:

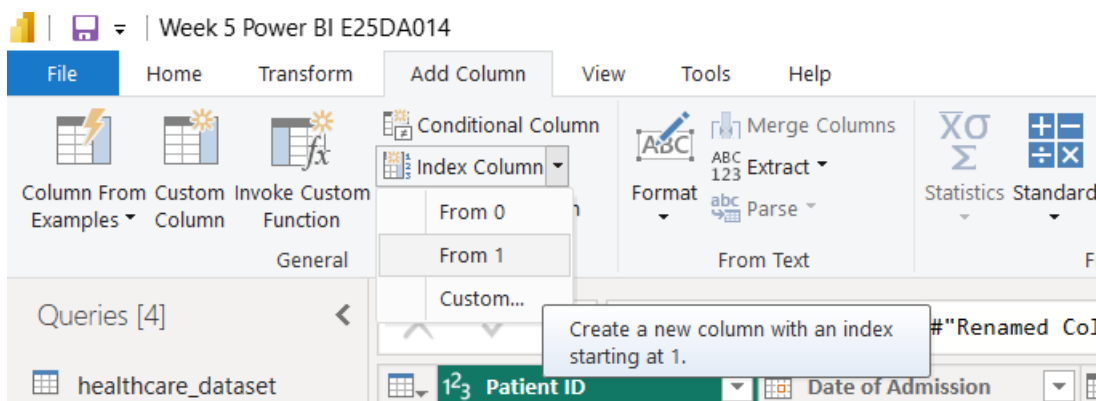
- Name
- Age
- Gender
- Blood Type
- Medical Condition

Table: SelectColumns(Source, ("Name", "Age", "Gender", "Blood Type", "Medical Condition"))

	Name	Age	Gender	Blood Type	Medical Condition
1	Bobby Jackson	30	Male	B-	Cancer
2	Leslie Terry	62	Male	A+	Obesity
3	Danny Smith	76	Female	A-	Obesity
4	Andrew Watts	28	Female	O+	Diabetes
5	Adrienne Bell	43	Female	AB+	Cancer
6	Emily Johnson	36	Male	A+	Asthma
7	Edward Edwards	21	Female	AB-	Diabetes
8	Christina Martinez	20	Female	A+	Cancer
9	Jasmine Aguilar	82	Male	AB+	Asthma
10	Christopher Berg	58	Female	AB-	Cancer
11	Michelle Daniels	72	Male	O+	Cancer
12	Aaron Martinez	38	Female	A-	Hypertension
13	Connor Hansen	75	Female	A+	Diabetes
14	Robert Bauer	68	Female	AB+	Asthma
15	Brooke Brady	44	Female	AB+	Cancer
16	Ms. Natalie Gamble	46	Female	AB-	Obesity
17	Haley Perkins	63	Female	A+	Arthritis
18	Mrs. Jamie Campbell	38	Male	AB-	Obesity
19	Luke Burgess	34	Female	A-	Hypertension
20	Daniel Schmidt	63	Male	B+	Asthma
21	Timothy Burns	67	Female	A-	Asthma
22	Christopher Bright	48	Male	B+	Asthma
23	Kathryn Stewart	58	Female	O+	Arthritis
24	Dr. Eileen Thompson	59	Male	A+	Asthma
25	Paul Henderson	72	Female	AB+	Hypertension
26	Peter Fitzgerald	73	Male	AB+	Obesity
27	Cathy Small	51	Female	O-	Asthma
28	Mr. Kenneth Moore	34	Female	A+	Diabetes
29	Mary Hunter	38	Female	O-	Hypertension
30	Joshua Oliver	63	Female	B+	Hypertension
31	Thomas Martinez	34	Male	B-	Asthma
32	James Patterson	23	Female	A+	Arthritis

Inappropriate or unnecessary columns were removed so that the table contained only relevant patient information.

In Dim_Patient table, Patient ID were created using Index Columns.



Go to :

Add Column and click Index Column from 1st Index.

Table.ReorderColumns("#Renamed Columns",{"Patient ID", "Name", "Age", "Gender", "Blood Type", "Age Group"})									
123 Patient ID	A6 Name	123 Age	A6 Gender	A6 Blood Type	A65 Age Group				
Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1	Bobby Jackson	30	Male	B-	19 - 35				
2	Leslie Terry	62	Male	A+	Above				
3	Danny Smith	76	Female	A-	Above				
4	Andrew Watts	28	Female	O+	19 - 35				
5	Adrienne Bell	43	Female	AB+	36 - 60				
6	Emily Johnson	36	Male	A+	36 - 60				
7	Edward Edwards	21	Female	AB-	19 - 35				
8	Christina Martinez	20	Female	A+	19 - 35				
9	Jasmine Aguilar	82	Male	AB+	Above				
10	Christopher Berg	58	Female	AB-	36 - 60				
11	Michelle Daniels	72	Male	O+	Above				
12	Aaron Martinez	38	Female	A-	36 - 60				
13	Connor Hansen	75	Female	A+	Above				
14	Robert Bauer	68	Female	AB+	Above				
15	Brooke Brady	44	Female	AB+	36 - 60				
16	Ms. Natalie Gamble	46	Female	AB-	36 - 60				
17	Haley Perkins	63	Female	A+	Above				
18	Mrs. Jamie Campbell	38	Male	AB-	36 - 60				
19	Luke Burgess	34	Female	A-	19 - 35				
20	Daniel Schmidt	63	Male	B+	Above				
21	Timothy Burns	67	Female	A-	Above				
22	Christopher Bright	48	Male	B+	36 - 60				
23	Kathryn Stewart	58	Female	O+	36 - 60				
24	Dr. Eileen Thompson	59	Male	A+	36 - 60				
25	Paul Henderson	72	Female	AB+	Above				
26	Cathy Small	51	Female	O-	36 - 60				
27	Mr. Kenneth Moore	34	Female	A+	19 - 35				
28	Mary Hunter	38	Female	O-	36 - 60				
29	Joshua Oliver	63	Female	B+	Above				
30	Thomas Martinez	34	Male	B-	19 - 35				
31	James Patterson	23	Female	A+	19 - 35				
32	William Cooper	78	Male	AB-	Above				

n profiling based on top 1000 rows

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Patient ID were added in Dim_Patient Table.

2. Dim Admission

The **Dim Admission** table contains information related to patient admissions.

Relevant fields include:

- Admission ID
- Date of Admission
- Doctor
- Hospital
- Admission Type
- Discharge Date
- Medication
- Test Results

Table: ReorderColumns(*Renamed Columns), ("Admission ID", "Medical Condition", "Admission Type", "Doctor", "Hospital", "Medication", "Room N")

	Admission ID	Medical Condition	Admission Type	Doctor	Hospital	Medication	Room N
1	1	Cancer	Urgent	Matthew Smith	Sons And Miller	Paracetamol	
2	2	Obesity	Emergency	Samantha Davies	Kim Inc	Ibuprofen	
3	3	Obesity	Emergency	Tiffany Mitchell	Cook Plc	Aspirin	
4	4	Diabetes	Elective	Kevin Wells	Hernandez Rogers And Vang,	Ibuprofen	
5	5	Cancer	Urgent	Kathleen Hanna	White-White	Penicillin	
6	6	Asthma	Urgent	Taylor Newton	Nunez-Humphrey	Ibuprofen	
7	7	Diabetes	Emergency	Kelly Olson	Group Middleton	Paracetamol	
8	8	Cancer	Emergency	Suzanne Thomas	Powell Robinson And Valdez,	Paracetamol	
9	9	Asthma	Elective	Daniel Ferguson	Sons Rich And	Aspirin	
10	10	Cancer	Elective	Heather Day	Padilla-Walker	Paracetamol	
11	11	Cancer	Urgent	John Duncan	Schaefer-Porter	Paracetamol	
12	12	Hypertension	Urgent	Douglas Mayo	Lyons-Blair	Lipitor	
13	13	Diabetes	Emergency	Kenneth Fletcher	Powers Miller, And Flores	Penicillin	
14	14	Asthma	Urgent	Theresa Freeman	Rivera-Gutierrez	Lipitor	
15	15	Cancer	Urgent	Roberta Stewart	Morris-Arellano	Paracetamol	
16	16	Obesity	Elective	Maria Dougherty	Cline-Williams	Aspirin	
17	17	Arthritis	Elective	Erica Spencer	Cervantes-Wells	Paracetamol	
18	18	Obesity	Urgent	Justin Kim	Torres, And Harrison Jones	Paracetamol	
19	19	Hypertension	Elective	Justin Moore Jr.	Houston Plc	Aspirin	
20	20	Asthma	Elective	Denise Galloway	Hammond Ltd	Penicillin	
21	21	Asthma	Elective	Krista Smith	Jones Llc	Aspirin	
22	22	Asthma	Urgent	Gregory Smith	Williams-Davis	Lipitor	
23	23	Arthritis	Urgent	Vanessa Newton	Clark-Mayo	Lipitor	
24	24	Asthma	Urgent	Donna Martinez Md	And Sons Smith	Lipitor	
25	25	Hypertension	Emergency	Stephanie Kramer	Wilson Group	Paracetamol	
26	26	Asthma	Elective	Wendy Glenn	Brown, And Jones Weaver	Ibuprofen	
27	27	Diabetes	Emergency	James Ellis	Serrano-Dixon	Lipitor	
28	28	Hypertension	Emergency	Jared Bruce Jr.	Gardner-Miller	Penicillin	
29	29	Hypertension	Elective	Brandy Mitchell	Guerrero-Boone	Paracetamol	
30	30	Asthma	Urgent	Jacob Huynh	Hart Ltd	Ibuprofen	
31	31	Arthritis	Urgent	Kristina Frazier	Cruz-Santiago	Penicillin	
32							

Query Settings

PROPERTIES

Name: Dim_Admission

APPLIED STEPS

Source

Removed Duplicates

Removed Other Columns

Added Index

Renamed Columns

Reordered Columns

1 profiling based on top 1000 rows

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Unnecessary columns were removed to keep the dimension table focused on admission-related information.

Also, In Dim_Admission table, Admission ID were created using Index Columns.

3. Fact Billing

The **Fact Billing** table was created to store billing-related information.

The required data was obtained from the healthcare dataset using **Merge Queries**.

Table: DistinctColumns(*Renamed Columns), ("Fact_billing", "Dim_Patient")

Merge

Select a table and matching columns to create a merged table.

Fact_billing

Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital
Bobby Jackson	30	Male	B-	Cancer	1/11/2024	Matthew Smith	Sons And Miller
Leslie Terry	62	Male	A+	Obesity	8/20/2019	Samantha Davies	Kim Inc
Danny Smith	76	Female	A-	Obesity	9/22/2022	Tiffany Mitchell	Cook Plc
Andrew Watts	28	Female	O+	Diabetes	11/28/2020	Kevin Wells	Hernandez Rogers And Vang,

Dim_Patient

Patient ID	Name	Age	Gender	Blood Type	Age Group
1	Bobby Jackson	30	Male	B-	19 - 35
2	Leslie Terry	62	Male	A+	Above
3	Danny Smith	76	Female	A-	Above
4	Andrew Watts	28	Female	O+	19 - 35
5	Adrienne Bell	43	Female	AB+	36 - 60

Join Kind: Left Outer (all from first, matching from second)

Use fuzzy matching to perform the merge

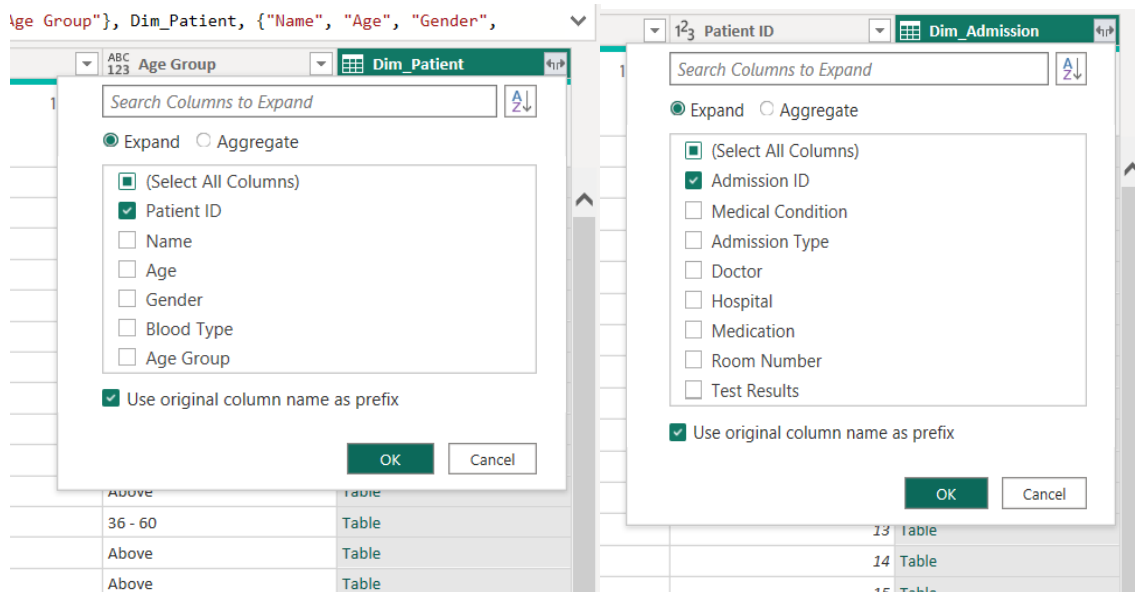
Fuzzy matching options

✓ The selection matches 1805 of 1805 rows from the first table.

OK Cancel

16 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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The necessary columns were merged and the identifiers were renamed as:

- Patient ID
- Admission ID

The billing-related information was retained for analysis.

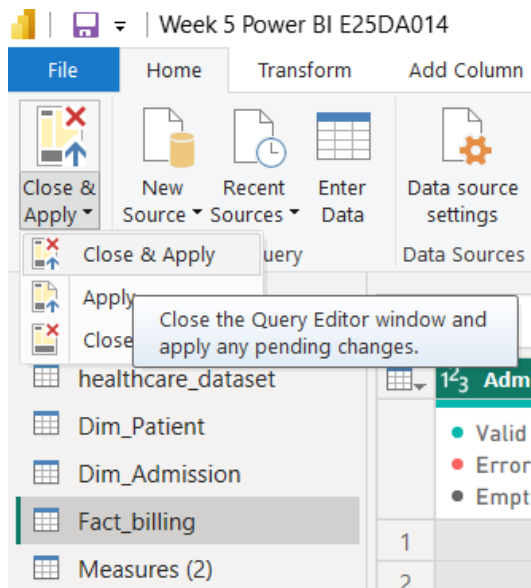
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6 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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	Date of Admission	Billing Amount	Discharge Date	Insurance Provider	Patient ID	Admission ID
1	1/31/2024	18856.28131	2/2/2024	Blue Cross	1	1
2	8/20/2019	33643.32729	8/26/2019	Medicare	2	2
3	9/22/2022	27955.09608	10/7/2022	Aetna	3	3
4	11/18/2020	37909.78241	12/18/2020	Medicare	4	4
5	9/19/2022	14238.31781	10/9/2022	Aetna	5	5
6	12/20/2023	48145.11095	12/24/2023	Unitedhealthcare	6	6
7	11/3/2020	19580.87234	11/15/2020	Medicare	7	7
8	12/18/2021	45820.46272	1/7/2022	Cigna	8	8
9	7/1/2020	50119.22279	7/14/2020	Cigna	9	9
10	5/23/2021	19784.63106	6/23/2021	Unitedhealthcare	10	10
11	4/19/2020	12576.79561	4/23/2020	Medicare	11	11
12	8/13/2023	7999.58668	9/5/2023	Medicare	12	12
13	12/12/2019	43282.28336	12/28/2019	Cigna	13	13
14	5/22/2020	33207.70663	6/19/2020	Unitedhealthcare	14	14
15	10/8/2021	40701.59923	10/13/2021	Unitedhealthcare	15	15
16	1/1/2023	12263.35743	1/11/2023	Blue Cross	16	16
17	6/23/2020	24499.8479	7/14/2020	Unitedhealthcare	17	17
18	3/8/2020	17440.46544	4/2/2020	Cigna	18	18
19	3/4/2021	18843.02302	3/14/2021	Blue Cross	19	19
20	11/15/2022	23762.20358	11/22/2022	Cigna	20	20
21	6/28/2023	42.51458855	7/2/2023	Blue Cross	21	21
22	1/21/2020	17695.91162	2/9/2020	Aetna	22	22
23	5/12/2022	5998.102908	6/10/2022	Aetna	23	23
24	8/2/2021	25250.05243	8/12/2021	Aetna	24	24
25	5/15/2020	33211.29542	6/8/2020	Medicare	25	25
26	12/23/2023	26786.52956	1/19/2024	Blue Cross	26	26
27	6/21/2022	18834.80134	6/30/2022	Unitedhealthcare	27	27
28	1/3/2021	32643.29935	1/16/2021	Cigna	28	28
29	10/9/2023	5767.011054	10/12/2023	Aetna	29	29
30	8/18/2019	47909.12881	9/1/2019	Cigna	30	30
31	11/3/2019	25835.32359	11/29/2019	Unitedhealthcare	31	31
32	5/18/2023	17993.2262	6/15/2023	Medicare	32	32

APPLIED STEPS: Source, Removed Duplicates, Merged Queries, Expanded Dim_Patient, Renamed Columns, Merged Queries1, Expanded Dim_Admission, Renamed Columns1, Removed Other Columns



After creating the required tables, the changes were loaded into Power BI using **Close & Apply**.

Creating Relationships:

Relationships were then created between the tables:

- **Dim Patient** → **Fact Billing** using **Patient ID**
- **Dim Admission** → **Fact Billing** using **Admission ID**

This created a structured data model where patient and admission information can be connected with billing information.

Data Model Structure:

The final model can be represented as:

Dim Patient

↓

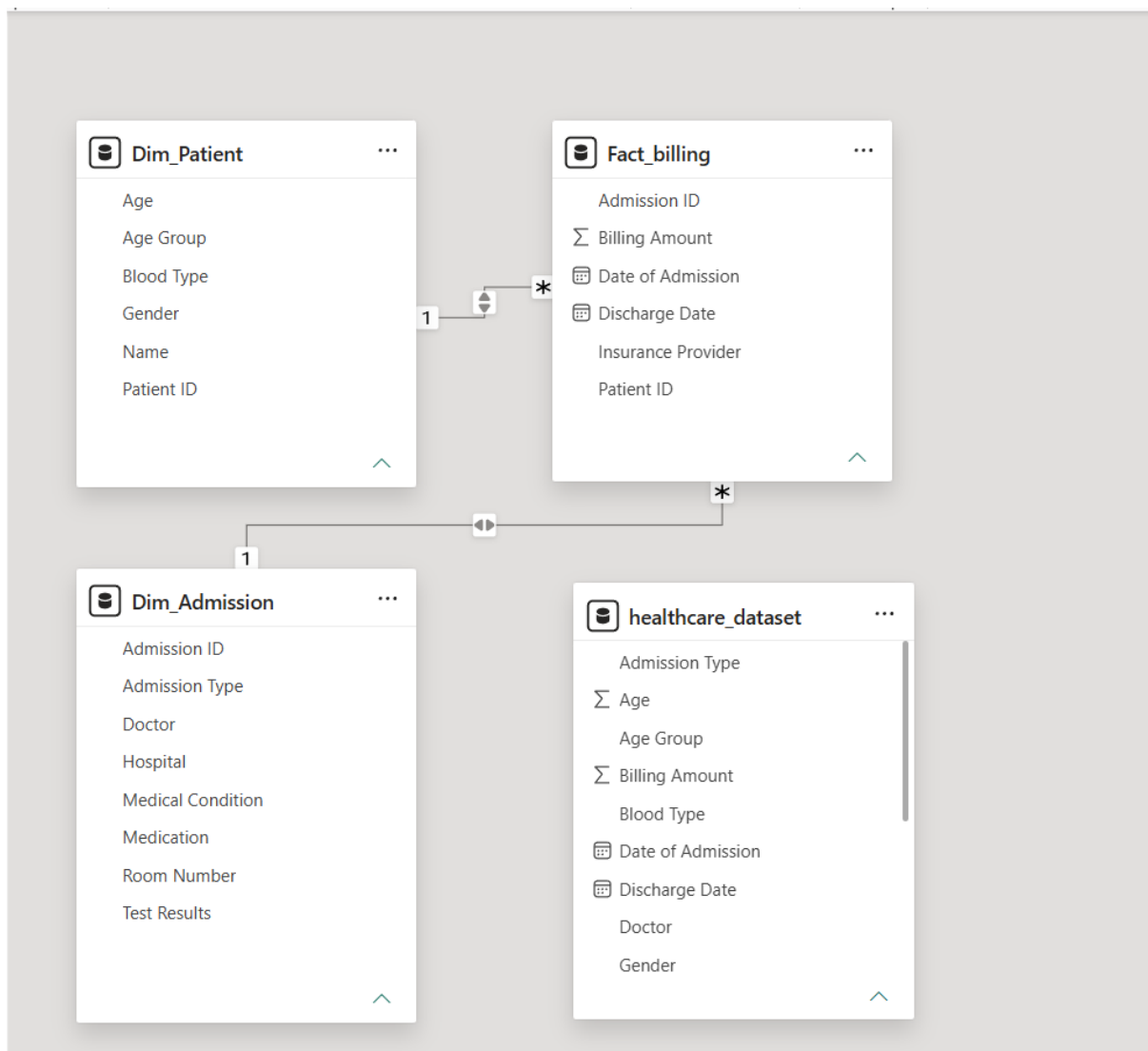
Fact Billing

↑

Dim Admission

The Fact Billing table acts as the central table, while Dim Patient and Dim Admission provide descriptive information for analysis.

Final Model View:



Result:

The healthcare dataset was successfully cleaned, transformed, and organized into a structured data model. The creation of **Dim Patient**, **Dim Admission**, and **Fact Billing** tables makes the data easier to analyze and supports efficient Power BI reporting.

Conclusion:

In Week 5, the healthcare dataset was prepared for analysis by performing data cleaning and data modeling operations. Text formatting, data type changes, and age-group classification were completed in Power Query. The dataset was then divided into dimension and fact tables, and relationships were established using Patient ID and Admission ID. This created a well-structured model suitable for further visualization and analysis in Power BI.